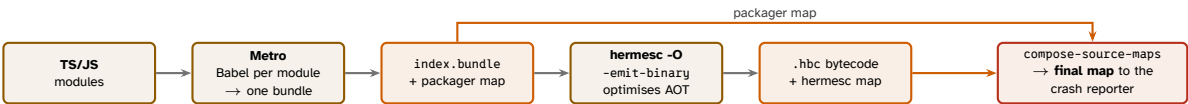
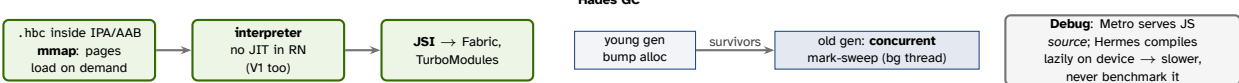


In one line: Hermes is Meta’s JS engine built for RN start-up: the release build **compiles JS to bytecode** (.hbc) with hermescc, so the phone **mmaps and interprets** it — no parse, no compile, small heap. Default since RN 0.70; the built-in JSC left core in **0.81**. **Hermes V1** — new compiler + VM — was opt-in in **0.82** and is the **default since 0.84**; it does *not* yet include native (“Static Hermes”) compilation or a JIT.

Release build (Gradle task / Xcode “Bundle React Native code and images”)



On the device



bytecode is tied to its runtime version (the Jun 2026 stable bumped it 98 -> 99); OTA bundles must be compiled by the same hermescc

How it works

- **AOT, not JIT:** parsing, scoping and optimisation happen on the build machine; the device only maps the file. Start-up (TTI) and memory win; peak CPU loses to a JIT engine — which an iOS app cannot use anyway (no writable+executable memory).
- The engine itself has more modes (AOT to bytecode or native, lazy compile, baseline JIT, mixable in one runtime); RN ships **bytecode + interpreter**.
- **Hades:** generational; the old generation is marked and swept **concurrently** on a background thread, so GC pauses stay short on the JS thread.
- **Is it Hermes?** `global.HermesInternal` — but that also exists when running *source*; prove .hbc in a release build.
- **Memory:** mapped bytecode is clean, file-backed memory — the OS can drop and re-read those pages; a parsed AST or JIT code would be dirty heap.

Hermes V1 — what changed

- “Next evolution”: improved **compiler and VM**. Versioned like 250829098.0.x (package hermes-compiler); legacy Hermes is 0.15.0.
- Measured on Expensify (0.82 post): bundle load **3.2%** faster on low-end Android, **9%** on iOS; total TTI 7.6% / 2.5%; content TTI 7.2% / 7.5%.
- **Not in V1 yet:** JS-to-native compilation (“Static Hermes”) and the JIT shown at RN EU 2023.
- Hermes stable 260318099.0.0 (Jun 2026): Set methods, `Object.groupBy`, iterator helpers, `Promise.withResolvers`, faster JSON; bytecode 99.

Opt in / opt out

RN	How
0.82-0.83 opt in	<code>hermesV1Enabled=true (gradle.properties)</code> · iOS <code>RCT_HERMES_V1_ENABLED=1</code> bundle exec pod install · pin hermes-compiler in resolutions/overrides; needs RN built from source (not prebuilt)
0.84+ opt out	overrides: <code>{"hermes-compiler": "0.15.0"}</code> · iOS <code>RCT_HERMES_V1_ENABLED=0</code> <code>RCT_USE_PREBUILT_RNCORE=0</code> · Android <code>hermesV1Enabled=false</code> + RN from source
JSC	0.79: moved to <code>@react-native-community/javascriptcore</code> ; 0.81: built-in removed

Example — what the release build does, by hand

```
npx react-native bundle --platform android --dev false \
  --entry-file index.js --bundle-output out/index.bundle \
  --sourcemap-output out/index.packager.map
hermescc -O -emit-binary -output-source-map \
  -out out/index.hbc out/index.bundle # + out/index.hbc.map
node node_modules/react-native/scripts/compose-source-maps.js \
  out/index.packager.map out/index.hbc.map -o out/index.map
// at runtime: which engine, which version?
const isHermes = () => !!global.HermesInternal;
HermesInternal.getRuntimeProperties()['OSS Release Version'];
```

Debugging and profiling

- **React Native DevTools** speaks CDP to Hermes *on the device*: breakpoints, console, heap snapshots; 0.83 added **Network** and **Performance** panels and a desktop app. (The old “debug in Chrome” ran JS in the browser — gone.)
- **Production stacks** are bytecode offsets (`index.android.bundle:1:48213`): upload the **composed** map per build *and* per OTA update (iOS: `SOURCEMAP_FILE` in the bundle phase (unverified)).
- **Profile a release-like build:** Android debugOptimized (0.82) keeps DevTools with C++ optimised — about 60 vs 20 FPS against debug.

Traps

- ! “Hermes V1 = Static Hermes” — no: V1 has neither native compilation nor the JIT.
- ! Benchmarking in debug measures lazy source compilation, not .hbc.
- ! A packager map alone does not decode Hermes stacks — compose it with hermescc’s.
- ! Upgrading to 0.81+ with JSC? Add the community package or the build loses it.
- ! Hermes needs no JIT to be fast at *start*; CPU-heavy loops are its weak spot — move them native (TurboModule) or to a worklet.

Remember

“**Compile on the Mac, map on the phone, compose the maps.**” V1: opt-in 0.82, default 0.84.